

# Schur dualities arising from quantum symmetric pairs

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## Abstract

A symmetric pair  $(\mathfrak{g}, \mathfrak{g}^\theta)$  consists of a Lie algebra  $\mathfrak{g}$  and a fixed point subalgebra  $\mathfrak{g}^\theta$ , where  $\theta$  is an involution on  $\mathfrak{g}$ . The corresponding quantum symmetric pair  $(U, U^\imath)$ , introduced by Letzter for finite type and Kolb for Kac-Moody type, is a  $q$ -deformation of  $(\mathfrak{g}, \mathfrak{g}^\theta)$ . Here  $U$  is the quantum group associated to  $\mathfrak{g}$  and  $U^\imath$  is a coideal subalgebra of  $U$ . The algebra  $U^\imath$ , now referred as the **iquantum group**, can be viewed as natural generalizations of quantum groups.

The Schur-Jimbo duality is one of the most fundamental topics in representation theory, bridging the irreducible representations of a Hecke algebra with those of a Drinfeld-Jimbo quantum group. In [1](#), the authors establish an  $\imath$ Schur duality where the Hecke algebra of type B, and the iquantum group of quasi-split type AIII satisfy a double centralizer property. Furthermore, it was shown that the Kazhdan-Lusztig basis of type B coincides with the  $\imath$ canonical bases arising from tensor product modules of iquantum groups.

We study iquantum groups of AIII in a more general sense and establish an  $\imath$ Schur duality which can be viewed as a common generalization of Jimbo-Schur duality and Bao-Wang's quasi-split  $\imath$ Schur duality. The  $\imath$ canonical bases in this setting provides a common generalization of both type A and type B Kazhdan-Lusztig bases.

**Theorem (2).** Let  $(U, U^\imath)$  be a quantum symmetric pair of type AIII. The iquantum group  $U^\imath$  and the Hecke algebra of type B form a double centralizer property on a tensor module of  $U$ .

The exploration of  $\imath$ Schur dualities within the framework of quantum symmetric pairs provides a natural extension of classical Schur dualities. Specifically, the type AI iquantum group serves as a quantization of the special orthogonal Lie algebra, while the type AII iquantum group serves as a quantization of the spin Lie algebra. These classical limits form Schur dualities with the Brauer algebra.

**Theorem (3).** Let  $(U, U')$  be a quantum symmetric pair of super type AI or AII. The  $q$ -quantum group  $U^i$  and the  $q$ -Brauer algebra form a double centralizer property on a tensor module of  $U$ .

Motivated by the immense success of quantum symmetric pairs, we introduce quantum supersymmetric pairs associated with Lie superalgebras in [4](#) and establish their fundamental properties. Furthermore, we extend the aforementioned  $i$ Schur dualities to the super setting.

## References

1. H. Bao and W. Wang, *A new approach to Kazhdan-Lusztig theory of type B via quantum symmetric pairs*, Astérisque **402** (2018), vii+134 pp.
2. Y. Shen and W. Wang,  *$i$ Schur duality and Kazhdan-Lusztig basis expanded*, Adv. Math. **427** (2023), Paper No. 109131.
3. W. Cui and Y. Shen, *Canonical basis of  $q$ -Brauer algebras and  $i$ Schur dualities*, Math Res. Let. (to appear).
4. Y. Shen, *Quantum supersymmetric pairs and  $i$ Schur duality of type AIII*, submitted.

## Original document

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